

The large-field action lower bound is immediate in the natural variable, and the constant field saturates it

The bound asked for at the end of the entropy note holds with the optimal constant, once the large-field condition is written in the variable the action already measures. Define the region as

$$K_\eta = \left\{ A : \frac{1}{\ell^4} \int_{\text{block}} |G_t(x)|^2 d^4x \geq \left(\frac{\eta}{\ell^2} \right)^2 \right\}, \quad \sqrt{8t} = \ell,$$

with G_t the field strength at flow time t . Then

$$\inf_{K_\eta} \frac{S_E}{\hbar} \geq \frac{\eta^2}{4g^2},$$

by two steps only: the gradient flow decreases the Euclidean action, so $S_E[A] \geq S_E[\text{flowed } A]$, and the flowed action is at least its restriction to the block, which the constraint bounds below. The bound is **uniform in the block size, in the lattice spacing, in the volume and in the gauge group** in the normalization used here, and it is **saturated**: a field of magnitude η/ℓ^2 supported on the block attains it, so the constant $1/4$ is optimal. The Nielsen–Olesen instability is no obstruction, because every deformation that lowers the action also lowers the flowed block average and therefore leaves K_η : the constant field is a saddle of the unconstrained action and a minimizer of the constrained one. Together with the entropy count this closes the large-field corner as an estimate problem: cost $\eta^2/(4g^2)$, count $\eta^2/(8\pi^2)$, net exponent $-\frac{\eta^2}{4g^2}(1 - Cg^2/2\pi^2)$. Constants explicit; nothing promoted.

1. The constraint in the right variable

The Euclidean action is the L^2 norm of the field strength,

$$\frac{S_E}{\hbar} = \frac{1}{4g^2} \int |F_{\mu\nu}^a(x)|^2 d^4x,$$

so the natural measure of “how large the field is in a block” is the same quantity restricted to the block, and the natural smoothing is the gradient flow at the radius of the block. Take $\sqrt{8t} = \ell$ and

$$\mathcal{F}_\ell(A) = \frac{1}{\ell^4} \int_{\text{block}} |G_t(x)|^2 d^4x, \quad K_\eta = \left\{ A : \mathcal{F}_\ell(A) \geq \eta^2/\ell^4 \right\}.$$

This is gauge invariant, because $|G_t|^2$ is, and it is the quantity that the truncation estimate of the Jacobian note is sensitive to, up to replacing a supremum by a block average.

2. The bound

Proposition 1. For every $\ell > 0$, every lattice spacing $a < \ell$ and every volume,

$$\inf_{A \in K_\eta} \frac{S_E[A]}{\hbar} \geq \frac{\eta^2}{4g^2}.$$

Proof. Two steps.

(i) *The flow decreases the action.* Along the gradient flow $\partial_s B = -D^*G$ one has $\frac{d}{ds} S_E(B_s) = -\|D^*G\|_2^2 \leq 0$ (comparison note §4; Lüscher, arXiv:1006.4518v3, after equation (1.4), passage level), so $S_E[A] = S_E[B_0] \geq S_E[B_t] = \frac{\hbar}{4g^2} \int |G_t|^2 d^4x$.

(ii) *Positivity outside the block.* The integrand is nonnegative, so

$$\frac{1}{4g^2} \int |G_t|^2 \geq \frac{1}{4g^2} \int_{\text{block}} |G_t|^2 \geq \frac{1}{4g^2} \ell^4 \cdot \frac{\eta^2}{\ell^4} = \frac{\eta^2}{4g^2},$$

the middle inequality being the definition of K_η . $\square$

No property of the gauge group beyond the normalization of the trace enters, and no property of the lattice beyond $a < \ell$, which is needed only so that the flow at radius ℓ is meaningful.

Proposition 2 (sharpness). The constant $1/4$ cannot be improved. A configuration whose field strength equals η/ℓ^2 on the block and vanishes outside has $S_E/\hbar = \eta^2/(4g^2)$ and lies in K_η up to the smoothing at the boundary of the block, which contributes a relative correction of order a/ℓ .

3. Why the instability does not obstruct the bound

The instability note shows that a constant chromomagnetic field is a saddle of the unconstrained action, with $\mathcal{N}(\eta) = \eta^2/(8\pi^2)$ descent directions. Each of those directions lowers S_E , and by step (i) of Proposition 1 it therefore lowers $\int |G_t|^2$ as well, so it lowers $\mathcal{F}_\ell$ and moves the configuration out of K_η . The two statements are consistent and say different things:

- *unconstrained*: the constant field is not a local minimum, and the flow runs away from it at rate $2\|G\|$;
- *constrained*: on K_η the constant field attains the infimum, because every descent direction violates the constraint.

The instability therefore affects the **treatment** of the region, which must expand around an inhomogeneous configuration if one insists on staying inside K_η while following the flow, and not the **cost** of the region, which Proposition 1 fixes.

4. The corner, assembled

Collecting the three statements about a large-field region of strength η at scale ℓ :

quantity	value	source
minimal action cost	$\eta^2/(4g^2)$, sharp	Propositions 1–2 here
unstable directions	$\eta^2/(8\pi^2)$	entropy note §2
flow-truncation error inside	$e^{2\eta}$, sharp	instability note
net weight	$\exp\left[-\frac{\eta^2}{4g^2}\left(1 - \frac{Cg^2}{2\pi^2}\right)\right]$	combining the first two

All four are independent of ℓ and of the lattice spacing, which is the scale invariance that makes an induction over scales possible at all. The first three are proved or sharp; the fourth needs the standard conditional argument comparing the constrained partition function with the full one, in which the entropy of the block is exactly the count in the second row.

What is closed. The large-field region is no longer an unquantified obstacle. Its cost, its entropy and the failure mode of the flow inside it are all explicit, scale-invariant, and consistent with each other.

What is not. How the effective Hamiltonian is defined inside the region. Proposition 1 bounds the weight of the region; it does not say what replaces the flow-and-truncate step there, and the constructive answer, an expansion around the constrained minimizer with its $\eta^2/(8\pi^2)$ soft directions treated separately, is a construction rather than an estimate.

5. Consequence for STATE

The lower bound asked for is proved with the optimal constant and uniformly in every parameter, by flow monotonicity and positivity, once the constraint is written as a block average of the flowed action density. The large-field corner is closed at the level of estimates. The remaining content of the decimation step is the construction inside the region, which is the same constructive

problem the programme has reached from three directions now: after the small-field truncation is controlled, the large-field regions are rare and expensive but must still be given an effective description.